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COMMON QUARTERLY EXAMINATION – 2026

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Standard XII

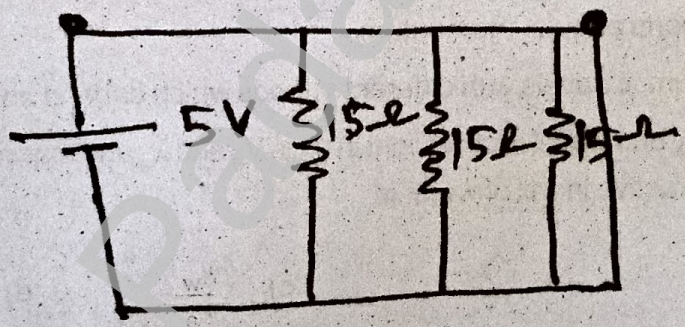
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PHYSICS

Time: 3.00 hrs. Part - I Marks: 70

I. Choose the correct answer: 15 x 1 = 15

1. Two metallic spheres of radii 1 cm and 3 cm are given charges of -1×10^{-2} C and 5×10^{-2} C respectively. If these are connected by a conducting wire, the final charge on the biggesr sphere is
   a) 3×10^{-2} C b) 4×10^{-2} C
   c) 1×10^{-2} C d) 2×10^{-2} C
2. A parallel plate capacitor filled with mica having $\varepsilon_r = 2$ is connected to a 10 V battery. The area of each parallel plate is 6 cm^2 and separation distance is 6 mm, then what is capacitance of capacitor?
   a) 0.77 PF b) 1.77 PF
   c) 17.7 PF d) 7.73 PF
3. What is the current drawn out from the battery?



   a) 1A b) 2A c) 3A d) 4A
4. The resistance of a uniform wire of length ℓ and cross-sectional area A is R. The resistance of wire of the same material having length 2ℓ and cross sectional area 2A is
   a) R b) 2R c) $R/2$ d) $R/4$
5. Three wires of equal lengths are bent in the form of loops. One of the loops is circle, another is a semi-circle and the third one is a square. They are placed in a uniform magnetic field and same electric current is passed through them. Which of the following loop configuration will experience greater torque?
   a) circle b) semi-circle
   c) square d) all of them

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6. The vertical component of Earth's magnetic field at a place is equal to the horizontal component. What is the value of angle of clip at this place?
   a) 30° b) 45° c) 60° d) 90°
7. An example of diamagnetic material is
   a) Nickel b) Water
   c) Aluminium d) Iron
8. An inductor 20 mH, a capacitor 50 μF and a resistor 40Ω are connected in series across a source of emf V = 10 sin 340 t. The power loss in AC circuit is
   a) 0.76 W b) 0.89 W c) 0.46 W d) 0.67 W
9. Q factor is equal to
   a) $\frac{\omega_r L}{R}$ b) $\frac{1}{R}\sqrt{\frac{L}{C}}$ d) $\frac{X_L}{R}$ d) all the above
10. Which of the following is an electromagnetic wave?
    a) α-rays b) β-rays c) γ-rays d) all of them
11. Which of the following is NOT true for electromagnetic waves?
    a) it transports energy
    b) it transports momentum
    c) it transports angular momentum
    d) In vacuum, it travels with different speeds which depend on their frequency
12. If the velocity and wavelength of light in air is V_a and λ_a and that in water is V_w and λ_w, then the refractive index of water is
    a) $\frac{V_w}{V_a}$ b) $\frac{V_a}{V_w}$ c) $\frac{\lambda_w}{\lambda_a}$ d) $\frac{V_a \lambda_a}{V_w \lambda_w}$
13. Stars twinkle due to
    a) reflection b) total internal reflection
    c) refraction d) polarisation
14. A plane glass is placed over a various coloured letters (violet, green, yellow, red). The letter which appears to be raised more is
    a) red b) yellow
    c) green d) violet
15. First diffraction minimum due to a single slit of width 1.0×10^{-5} cm is at 30°. Then wavelength of light used is
    a) 400 A° b) 500 A° c) 600 A° d) 700 A°

Part - II

II. Answer any 6 questions. **6 x 2 = 12**

(Q.No.24 is compulsory)

16. What is corona discharge?
17. State Kirchhoff's voltage rule.
18. Is an ammeter connected in series or parallel in a circuit? Why?
19. The repulsive force between two magnetic poles in air is 9×10^{-3} N. If the two poles are equal in strength and are separated by a distance of 10 cm, calculate the pole strength of each pole.
20. State Lenz's law.
21. What is displacement current?
22. Light travels from air into a glass slab of thickness 50 cm and refractive index 1.5, what is the speed of light in the glass slab?
23. Write down conditions for obtaining clear and broad interference fringes by Young's double slit experiment.
24. If the focal length of lens is 150 cm, what is the power of the lens?

Part - III

III. Answer any 6 questions. **6 x 3 = 18**

(Q.No.33 is compulsory)

25. Differentiate coulomb force and gravitational force.
26. Write the properties of paramagnetic materials.
27. How will you induce an emf by changing the area enclosed by the coil?
28. What are Fraunhofer lines? How are they useful in the identification of elements present in the sun?
29. Derive the relation between f and R for a spherical mirror.
30. Find the ratio of the intensities of light with wavelength 500 nm and 300 nm which undergo Rayleigh scattering.
31. Mention the differences between interference and diffraction.
32. Two light sources of equal amplitudes interfere interfere with each other. Calculate the ratio of maximum and minimum intensities.
33. The resistance of a nichrome wire at 20°C is 10Ω. If its temperature coefficient of resistivity of nichrome is 0.004/°C. Find the resistance of the wire at boiling point of water. Comment on the result.

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Part - IV

IV. Answer all the questions. 5 x 5 = 25

34. a) State Gauss law. Obtain the expression for electric field due to an Infinitely long charged wire.

(OR)

b) Derive the expression for the force between two parallel, current carrying conductors.

35. a) Obtain the condition for bridge balance in Wheatstone's bridge.

(OR)

b) Explain the working of single phase AC generator with neat diagram.

36. a) Explain the principle, construction and working of transformer.

(OR)

b) Write down Maxwell equations in integral form.

37. a) What is emission spectrum? Explain the types of emission spectrum.

(OR)

b) Determine the internal resistance of a cell using voltmeter.

38. a) Derive an expression for electric field due to a dipole at a point on its axial line.

(OR)

b) Derive the mirror equation and the equation for lateral magnification.

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